

brain-drain 4-bay disk sanitizer carrier, CM5

power-input

12 V DIN input, fuse, TVS, reverse-polarity FET, bulk capacitance

File: sheets/power-input.kicad_sch

power-bucks

5V_SYS and 5V_HDD (TPS56637), +3V3 and 3V3_M2 (AP63203), +1V2 LDO

File: sheets/power-bucks.kicad_sch

cm5

Compute Module 5 (wireless): power, control, GPIO, microSD, USB-C, UART, RTC cell, fan, DIP, LEDs, OLED

File: sheets/cm5.kicad_sch

usb3-hub-B

USB5744 hub B on CM5 USB3-1, bays 3 and 4

File: sheets/usb3-hub-B.kicad_sch

bridge-1

Bay 1: ASM1153E USB-SATA bridge and 22-pin SATA receptacle

File: sheets/bridge-1.kicad_sch

bay-switch-1

Bay 1: switched 12 V and 5 V with soft-start and PTC fuses

File: sheets/bay-switch-1.kicad_sch

bay-switch-2

Bay 2: switched 12 V and 5 V with soft-start and PTC fuses

File: sheets/bay-switch-2.kicad_sch

bridge-3

Bay 3: ASM1153E USB-SATA bridge and 22-pin SATA receptacle

File: sheets/bridge-3.kicad_sch

bay-switch-3

Bay 3: switched 12 V and 5 V with soft-start and PTC fuses

File: sheets/bay-switch-3.kicad_sch

bay-switch-4

Bay 4: switched 12 V and 5 V with soft-start and PTC fuses

File: sheets/bay-switch-4.kicad_sch

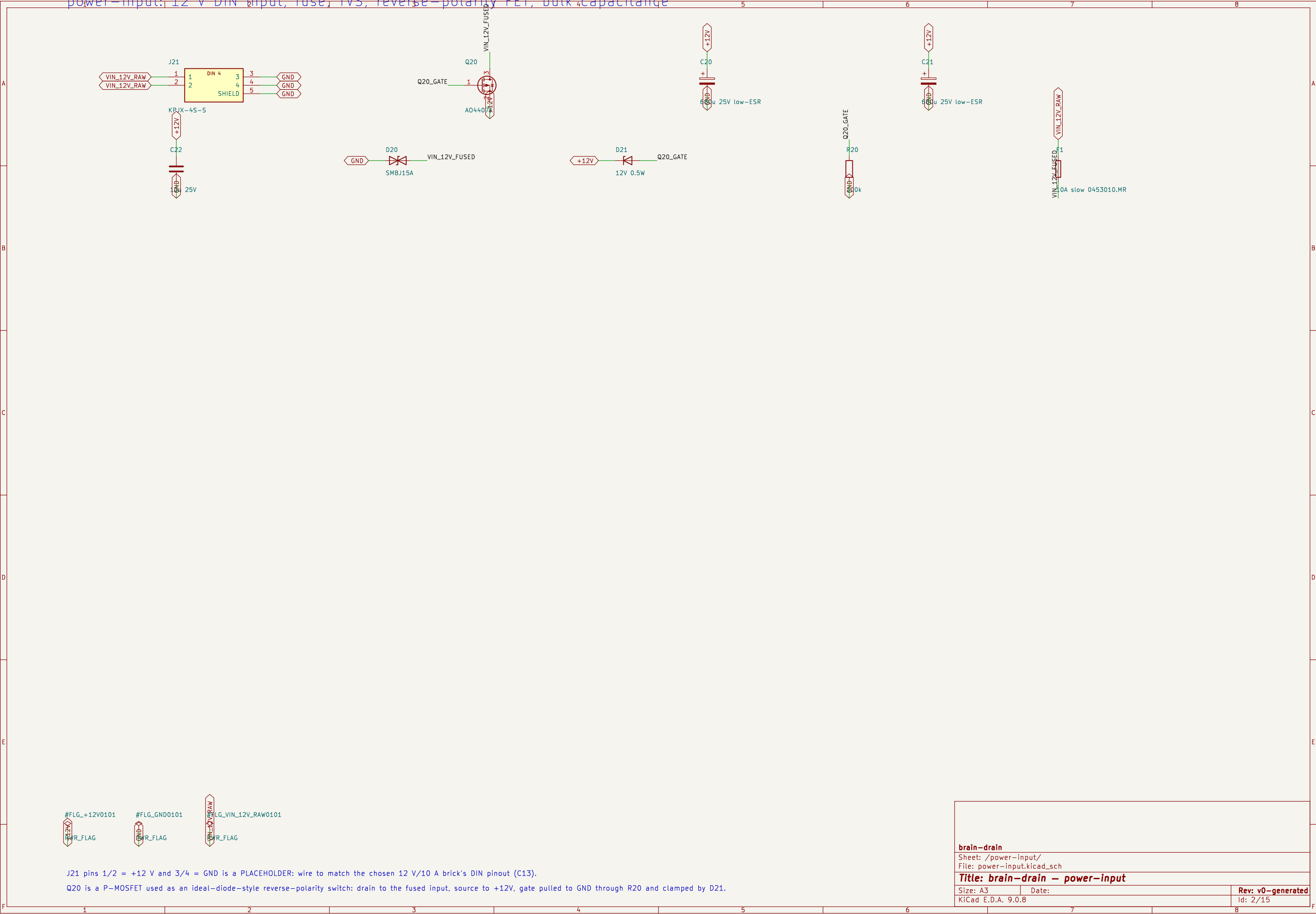
m2-nvme

Bay 5: M.2 M-key on PCIe Gen3 x1, 3V3_M2 buck + load switch, door switch

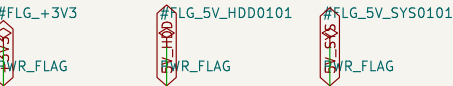
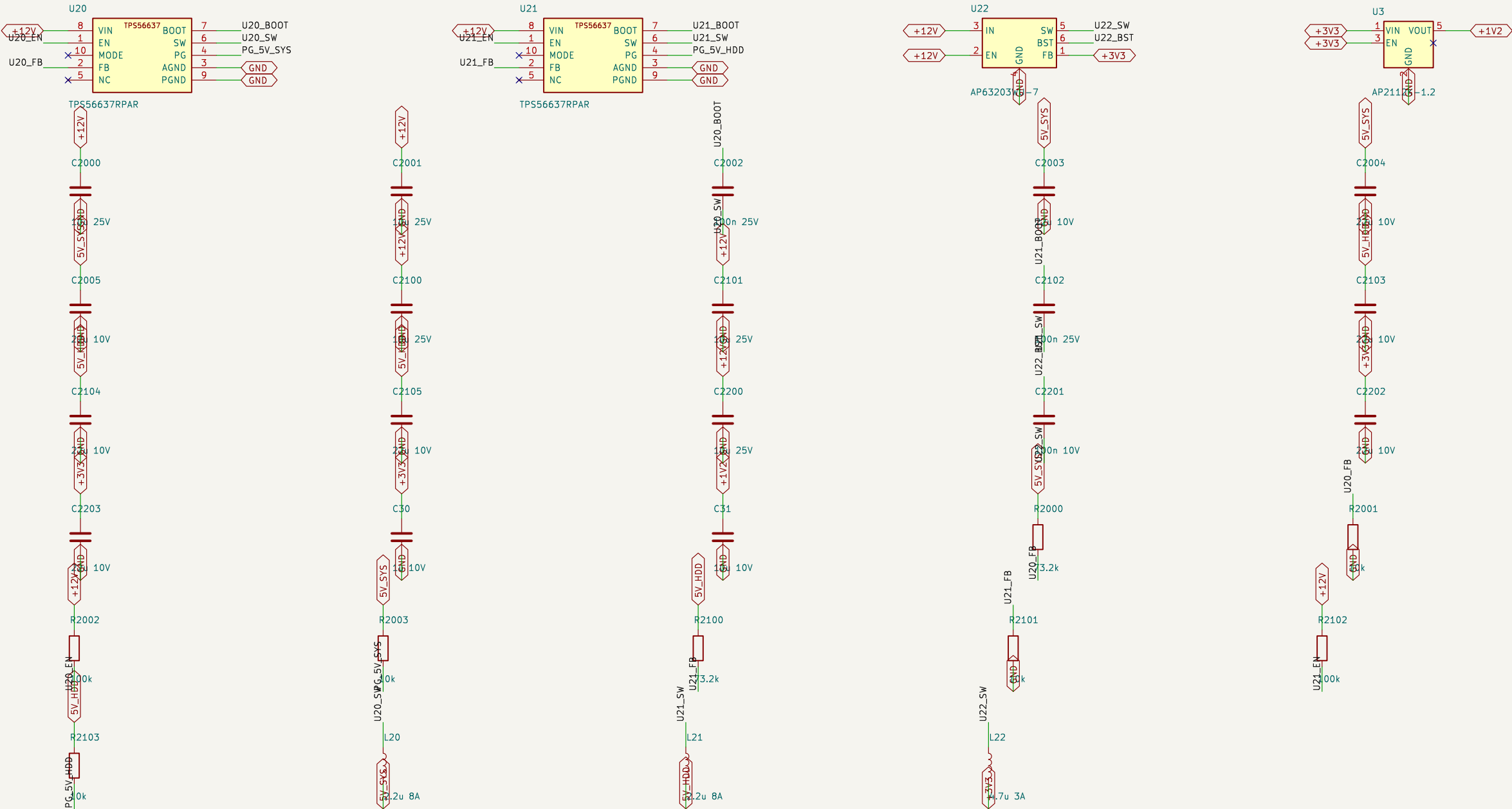
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brain-drain		
Sheet: /		
File: brain-drain.kicad_sch		
Title: brain-drain 4-bay disk sanitizer carrier, CM5		
Size: A3	Date:	Rev: v0-generated
KiCad E.D.A. 9.0.8	Id: 1/15	

power-input: 12 V DIN input, fuse, TVS, reverse-polarity FET, bulk capacitance

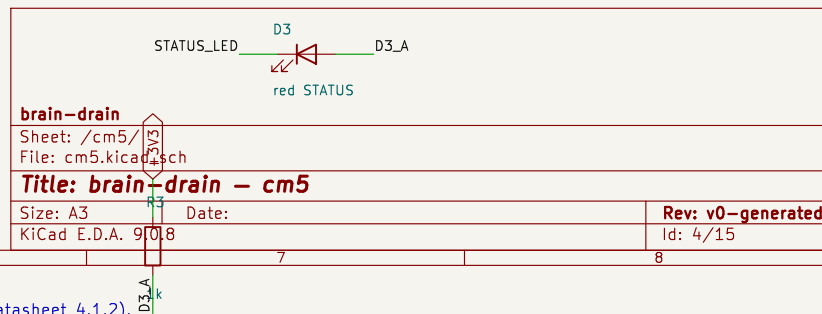
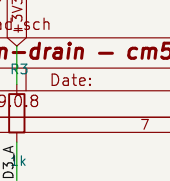
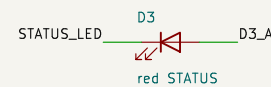
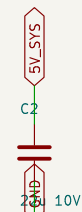
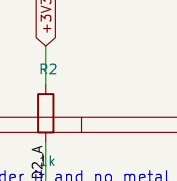
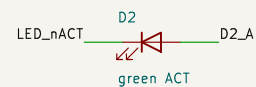
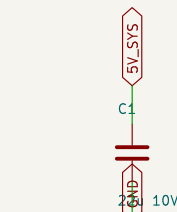
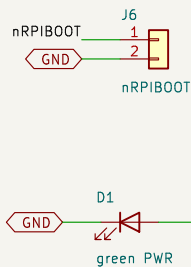
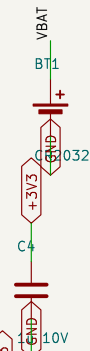
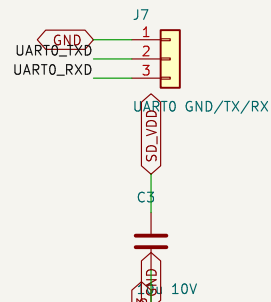
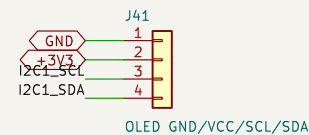
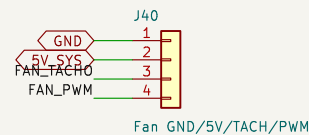
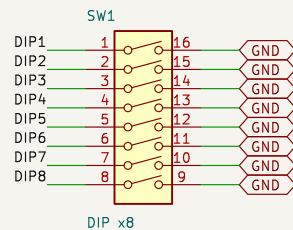
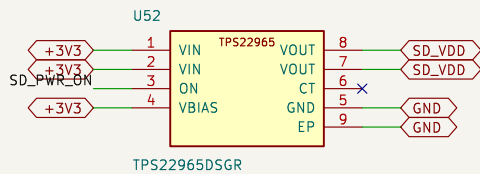
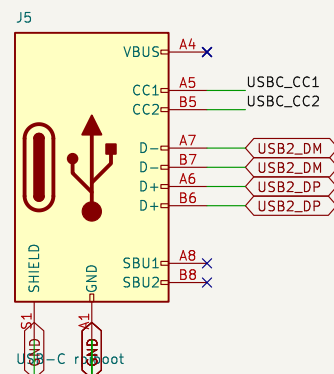
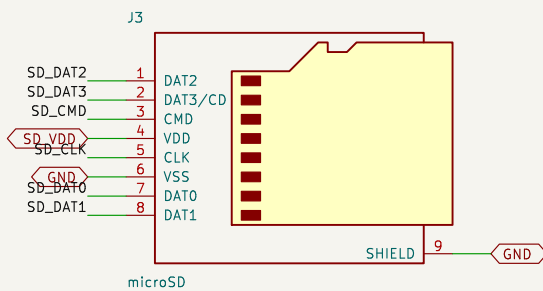
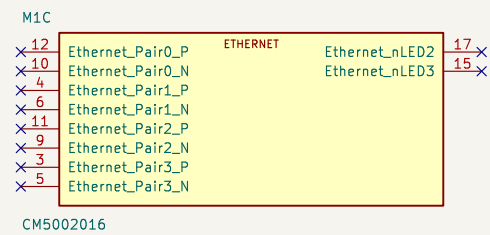
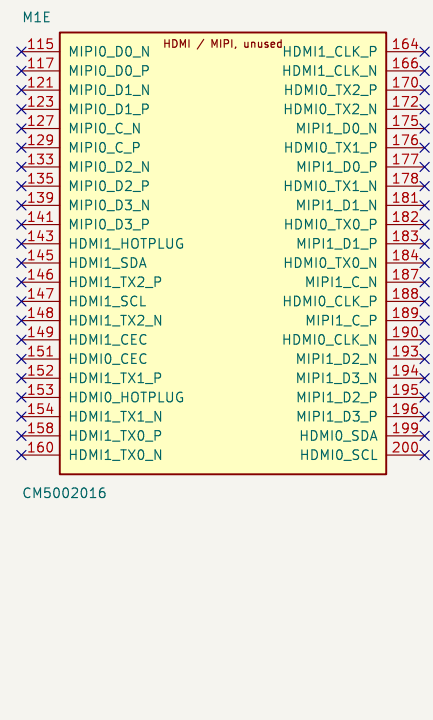
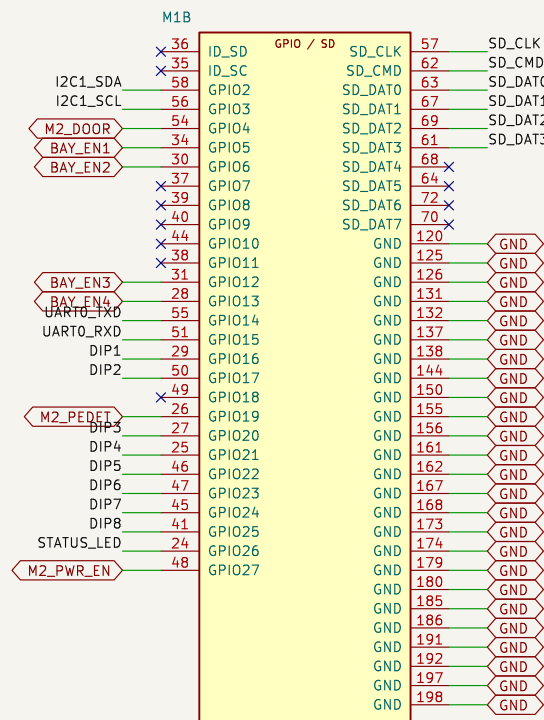
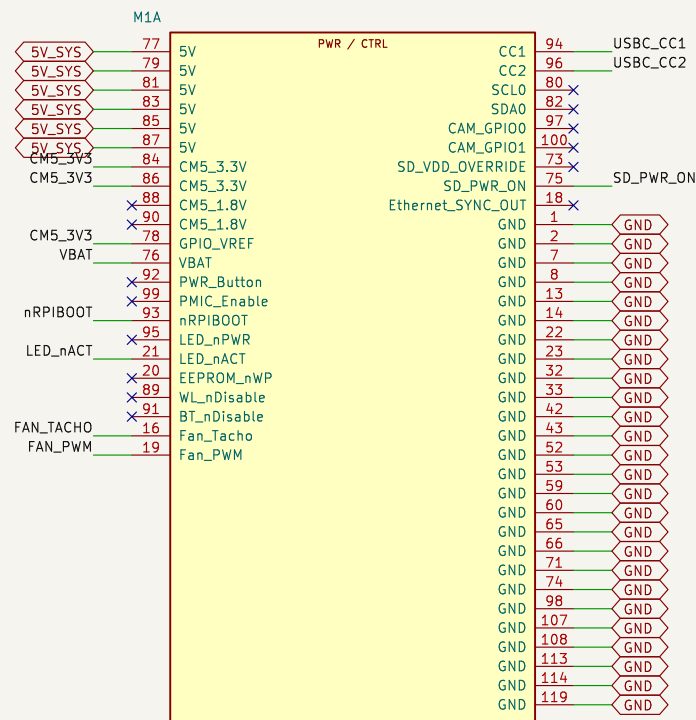


power-bucks: 5V_SYS and 5V_HDD (TPS56637), +3V3 and 3V3_M2 (AP63203), +1V2 LDO



TPS56637 FB: Vref 0.6 V; 73.2k/10k gives 4.99 V. MODE floating = forced CCM (quietest for the CM5 rail).
AP63203WU is the fixed 3.3 V variant: FB ties straight to the output. EN tied to VIN (always on).

cm5: Compute Module 5 (wireless): power, control, GPIO, microSD, USB-C, UART, RTC cell, fan, DIP, LEDs, OLED



GPIO_VREF is tied to the CM5's own 3.3 V output (pins 84/86) for 3.3 V GPIO signalling, per datasheet §4.2.

PMIC_Enable and PWR_Button are left floating (internal pull-ups). nRPIBOOT goes to jumper J6.

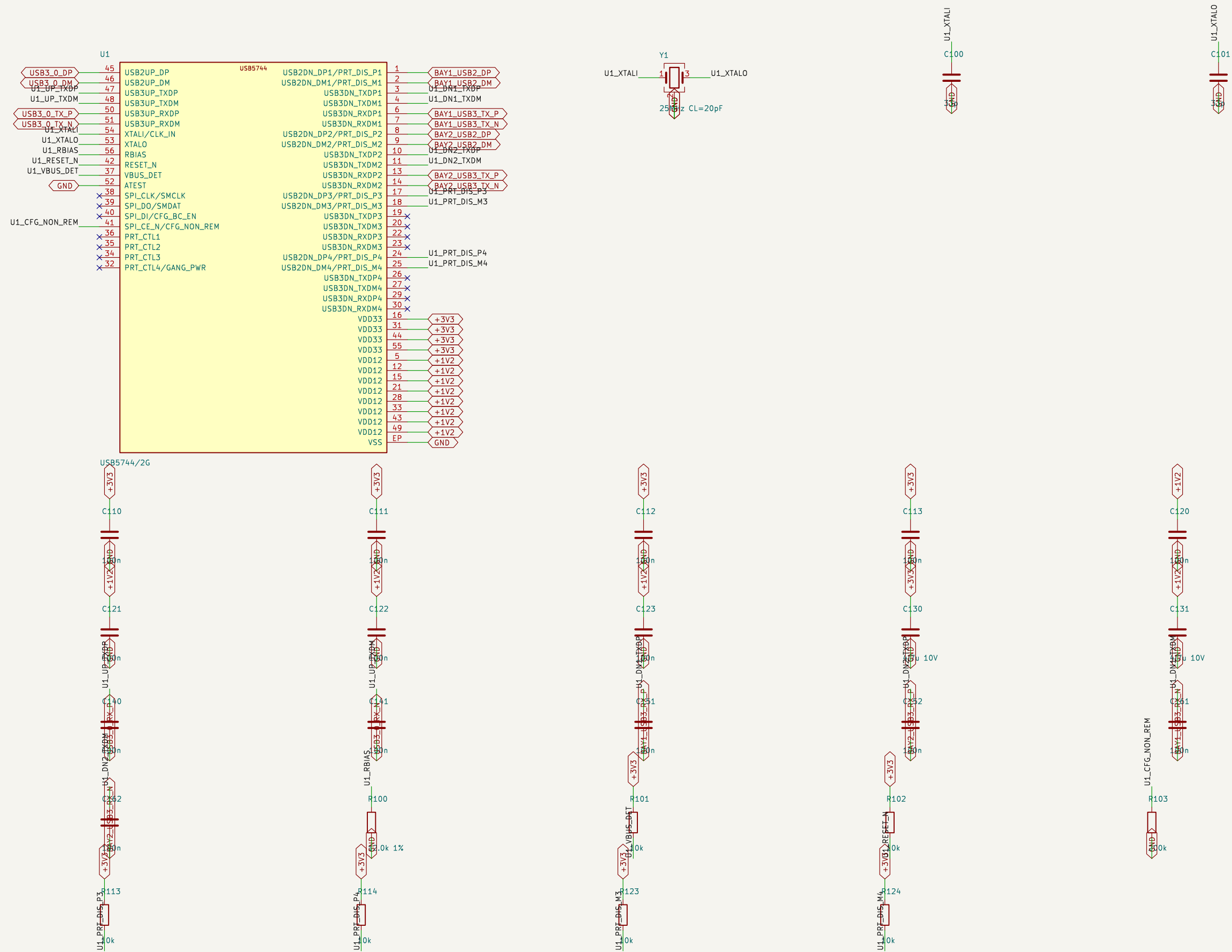
LED_nACT is an open-drain 20 mA output on the CM5; it sinks D2 directly. GPIO26 sinks D3 (software drives it low to light).

microSD power goes through U52 so the CM5 can power-cycle the card on reboot via SD_PWR_ON, matching the CM5IO reference (which uses an RT9742).

Ethernet dropped (C22). The CM5 wireless variant's PCB antenna is on the short module edge that carries MH1; that edge sits on the board's left edge with an 8 mm copper-free strip under it and no metal within 10 mm (CM5 datasheet 4.1.2).

USB-C is a design requirement (with a caveat). Across the CM5IO, CC1/CC2 connects to itself. VBUS is not connected (the board is powered from 12 V). USB_OTG_ID (status indicator)

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usb3-hub-A: USB5744 hub A on CM5 USB3-0, bays 1 and 2
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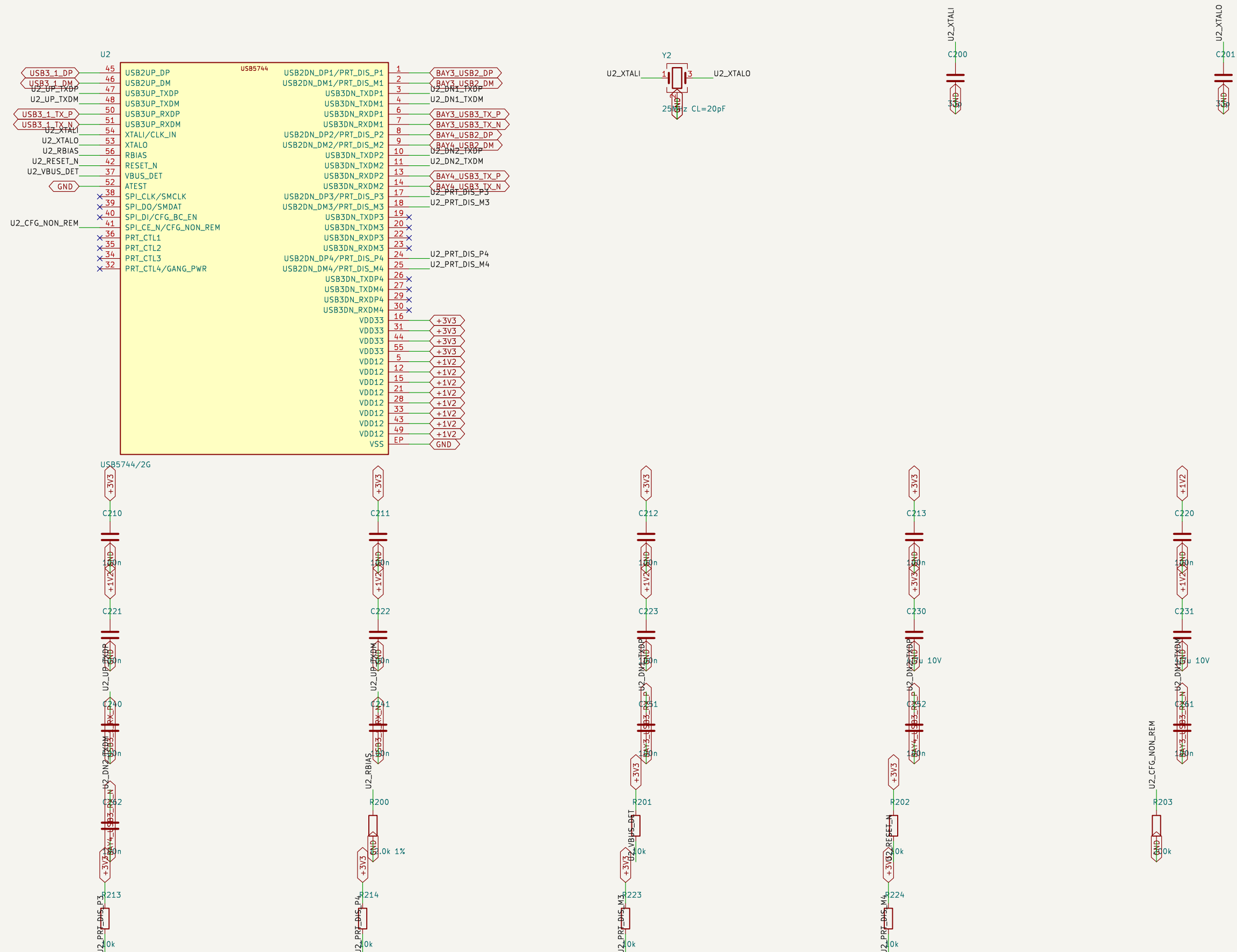


Hub A: upstream on CM5 USB3-0. Downstream ports 1,2 -> bays 1,2. Ports 3,4 disabled by PRT_DIS straps (verify the strap resistor value against USB5744 §3.4.2).

No SPI ROM and no SMBus pull-ups: the hub runs its internal ROM defaults. RESET_N has a pull-up only; add an RC if strap timing (1 ms hold) is a concern.

Crystal: 25 MHz, CL 20 pF per datasheet table 10-10; 33 pF load caps assume ~3 pF stray, adjust to the chosen crystal.

usb3-hub-B; USB5744 hub B on CM5 USB3-1, bays 3 and 4

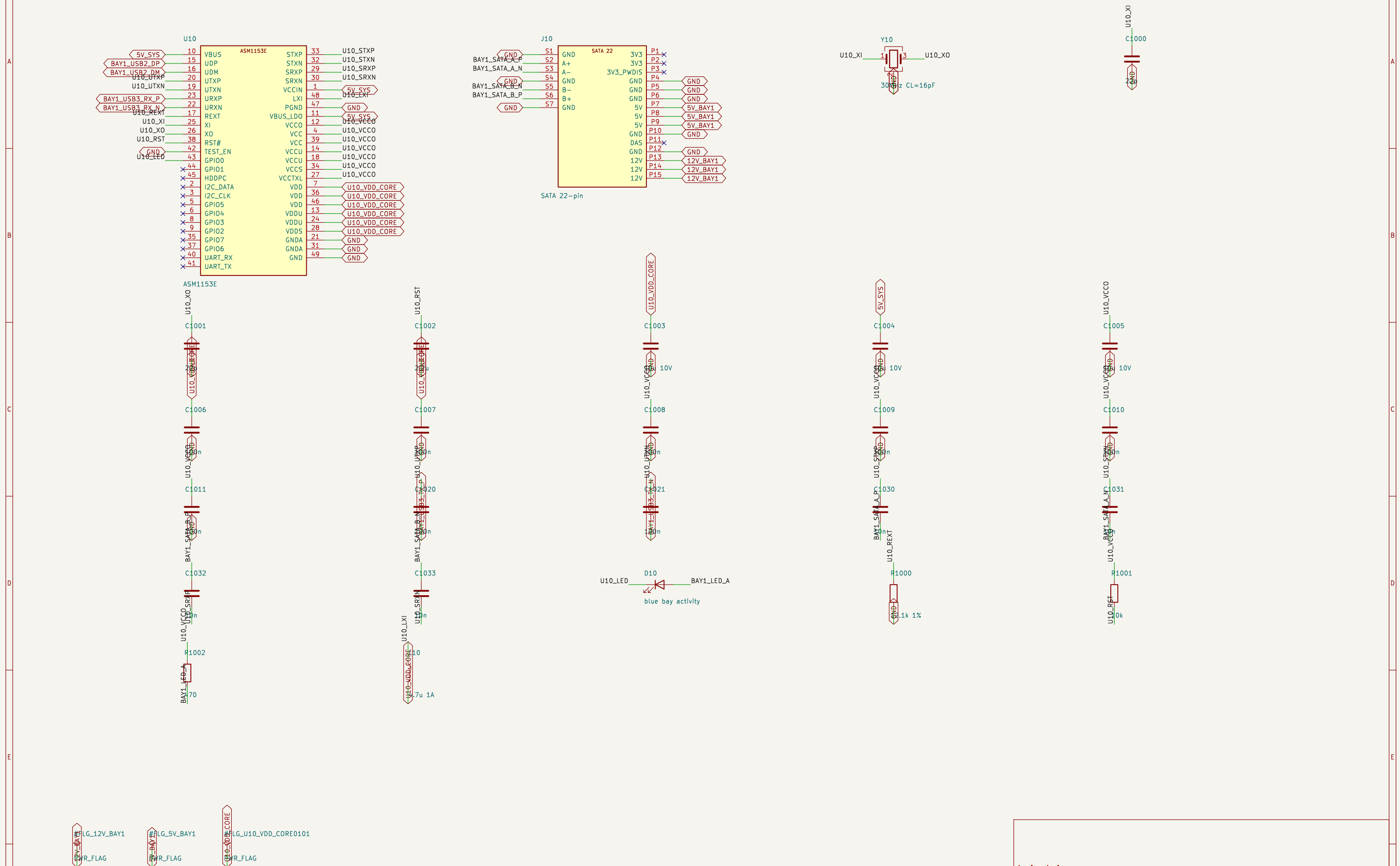


Hub B: upstream on CM5 USB3-1. Downstream ports 1,2 -> bays 3,4. Ports 3,4 disabled by PRT_DIS straps (verify the strap resistor value against USB5744 §3.4.2).

No SPI ROM and no SMBus pull-ups: the hub runs its internal ROM defaults. RESET_N has a pull-up only; add an RC if strap timing (1 ms hold) is a concern.

Crystal: 25 MHz, CL 20 pF per datasheet table 10-10; 33 pF load caps assume ~3 pF stray, adjust to the chosen crystal.

bridge-1: Bay 1: ASM1153E USB-SATA bridge and 22-pin SATA receptacle



ASM1153E runs from 5 V only: VBUS_LDO -> VCCO (3.3 V, feeds VCC/VCCU/VCCS/VCCTXL) and VCCIN -> LXI switcher -> 1.05 V core (VDD/VDDU/VDDS). Datasheet notes VCCIN must be tied to 5 V.

Clock straps GPIO3/GPIO7 left at their internal pull-up default (11) = 30 MHz crystal, so no strap resistors. GPIO6 default = I2C mode, no SPI ROM.

SATA P1–P3 (3.3 V) are not connected on purpose: P3 is PWDIS on SATA 3.3 drives and a 3.3 V supply there keeps them from spinning up.

Bay activity LED assumed on GPIO0 (bridge firmware default): confirm on the bench; GPIO1 is the alternative.

brain-drain

Sheet: /bridge-1/
File: bridge-1.kicad_sch

Title: *brain-drain – bridge-1*

Size: A3

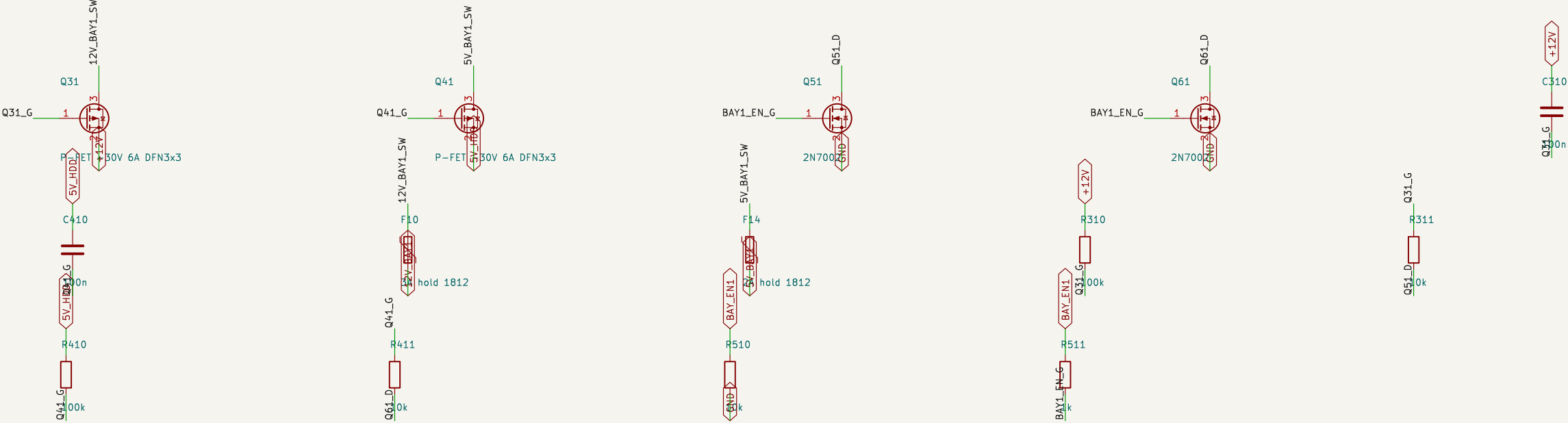
Date:

Rev: v0-generated

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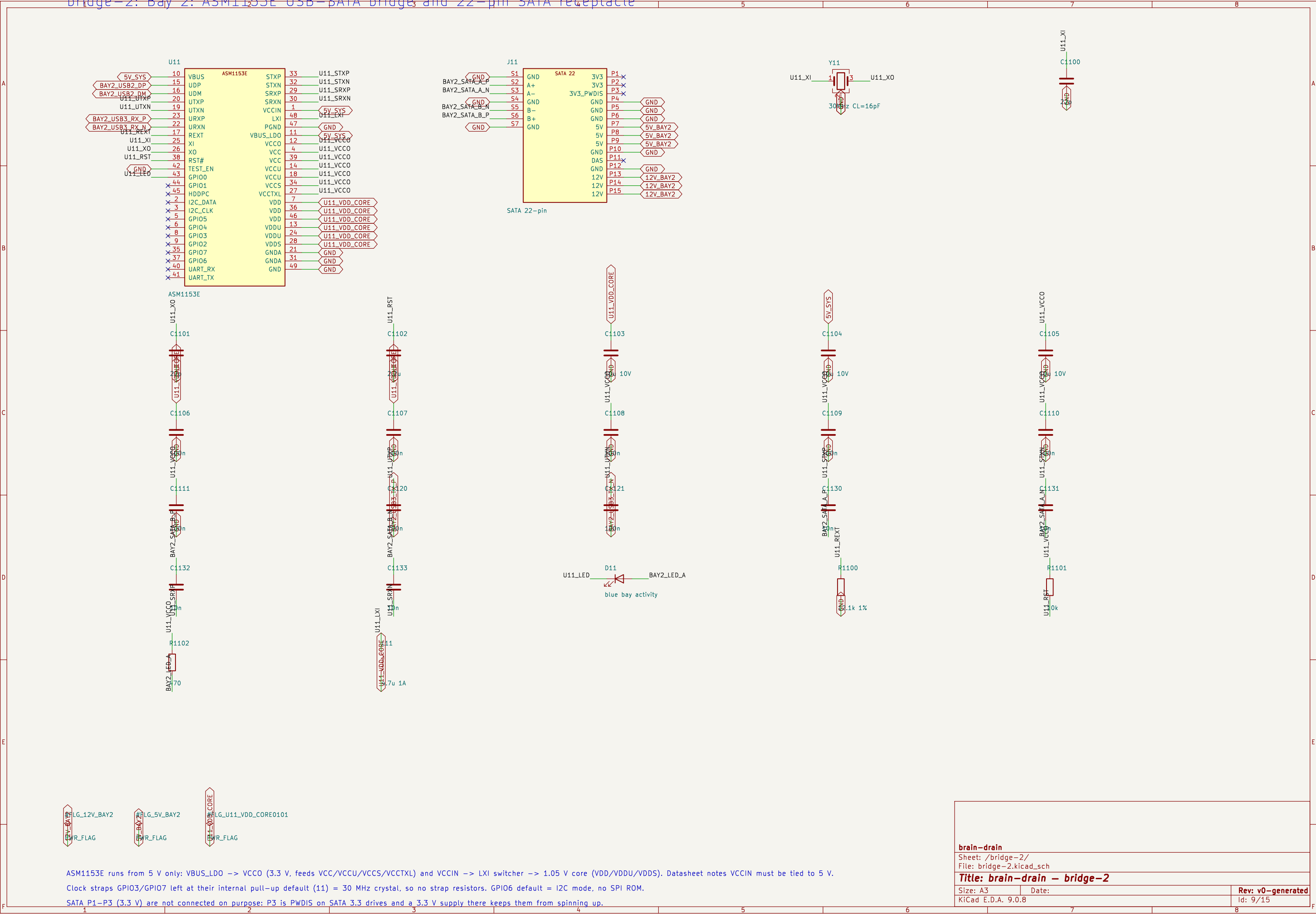
Id: 7/15

bay-switch-1: Bay 1: switched 12 V and 5 V with soft start and PTC fuses

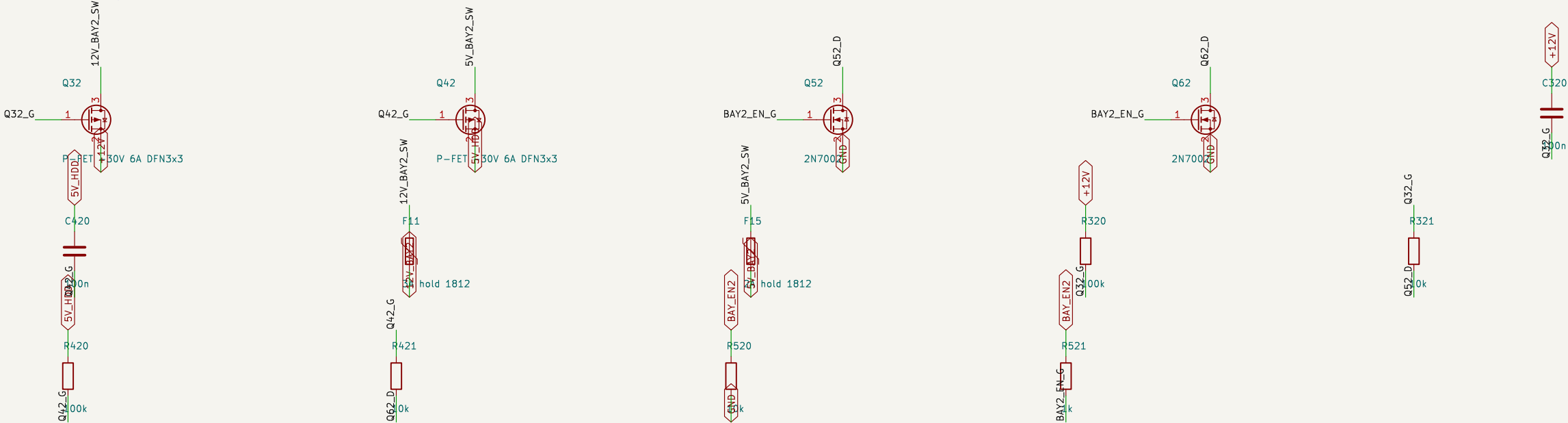


P-FET high-side switch; 100k gate pull-up holds it off; the 2N7002 pulls the gate down through 10k with 100n gate-source, ~1 ms ramp so the drive's bulk caps do not trip the fuse. BAY_EN pull-down keeps the bay off through bay

bridge-2: Bay 2: ASM1153E USB-SATA bridge and 22-pin SATA receptacle



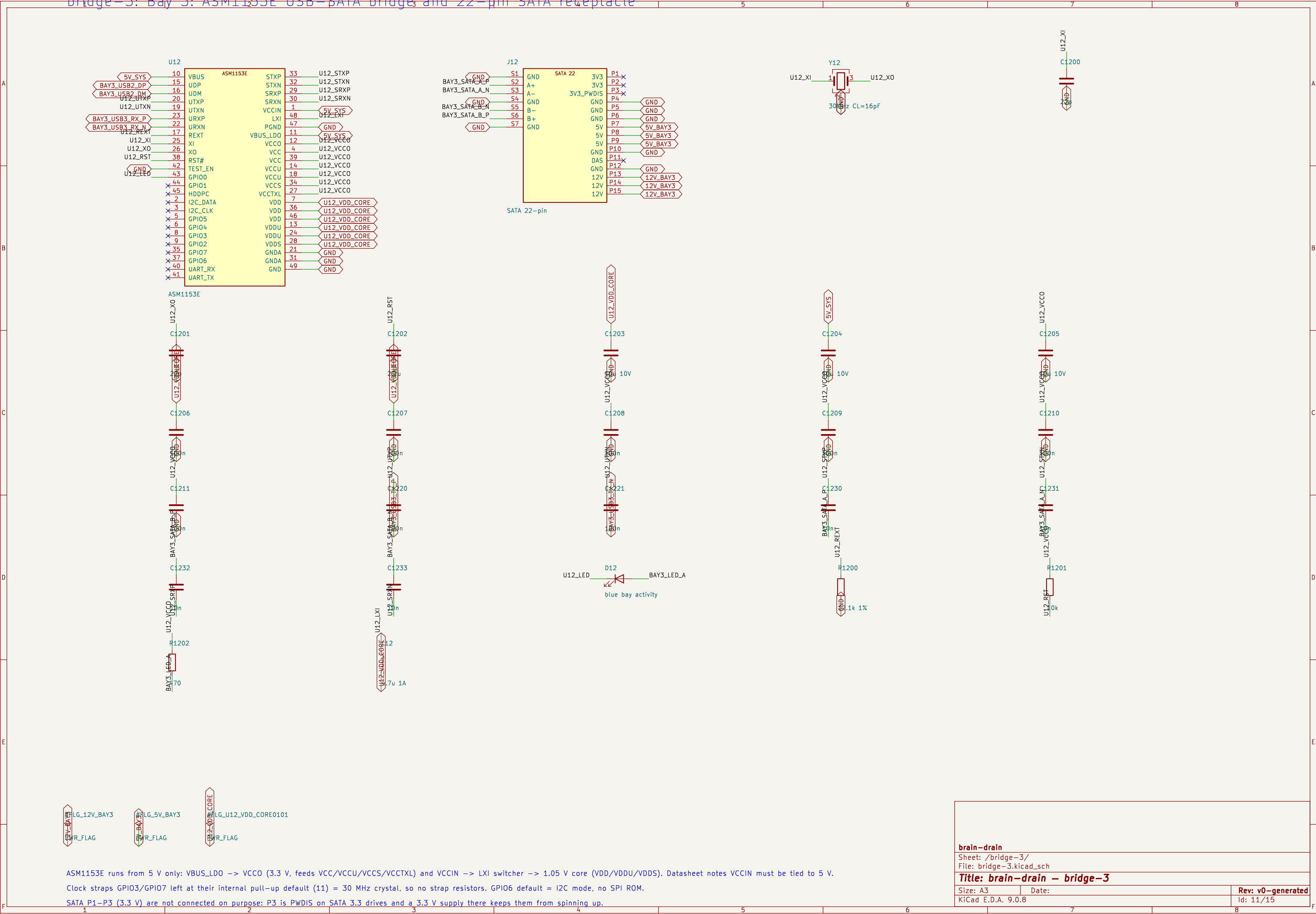
bay-switch-2: Bay 2: switched 12 V and 5 V with soft-start and PTC fuses



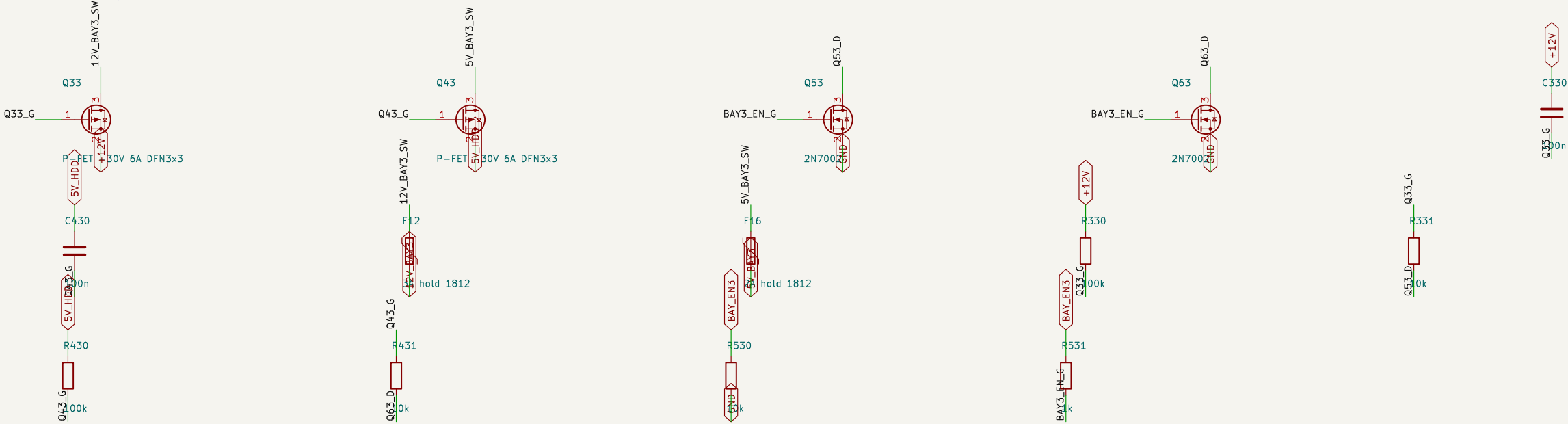
P-FET high-side switch; 100k gate pull-up holds it off; the 2N7002 pulls the gate down through 10k with 100n gate-source, ~1 ms ramp so the drive's bulk caps do not trip the fuse. BAY_EN pull-down keeps the bay off through bay

brain-drain		
Sheet: /bay-switch-2/ File: bay-switch-2.kicad_sch		
Title: brain-drain - bay-switch-2		
Size: A3	Date:	Rev: v0-generated
KiCad E.D.A. 9.0.8		Id: 10/15

bridge-3: Bay 3: ASM1153E USB-SATA bridge and 22-pin SATA receptacle



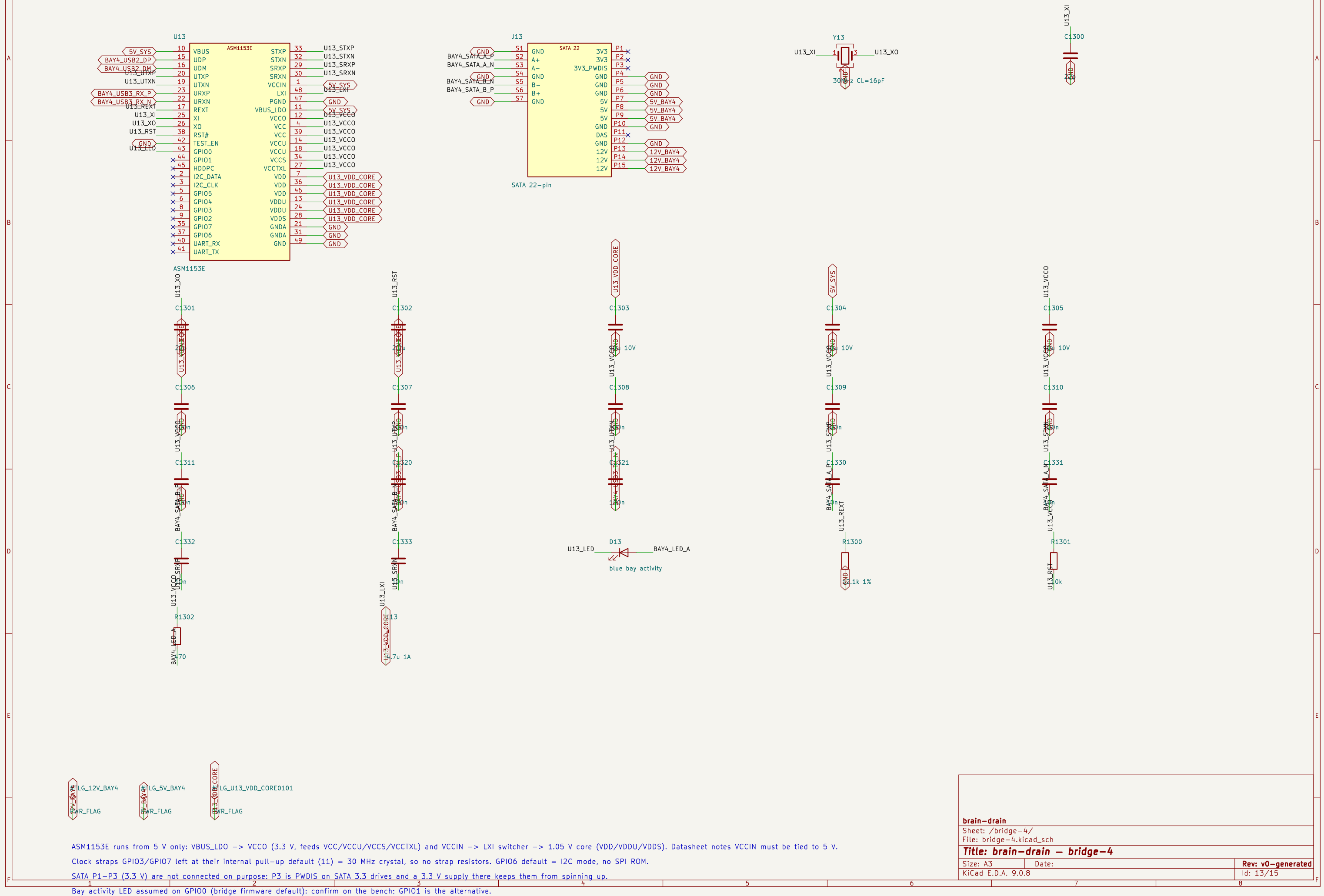
bay-switch-3: Bay 3: switched 12 V and 5 V with soft-start and PTC fuses



P-FET high-side switch; 100k gate pull-up holds it off; the 2N7002 pulls the gate down through 10k with 100n gate-source, ~1 ms ramp so the drive's bulk caps do not trip the fuse. BAY_EN pull-down keeps the bay off through bay

brain-drain		
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Title: brain-drain - bay-switch-3		
Size: A3	Date:	Rev: v0-generated
KiCad E.D.A. 9.0.8		Id: 12/15

bridge-4: Bay 4: ASM1153E USB-SATA bridge and 22-pin SATA receptacle



ASM1153E runs from 5 V only: VBUS_LDO -> VCCO (3.3 V, feeds VCC/VCCU/VCCS/VCCCTL) and VCCIN -> LXI switcher -> 1.05 V core (VDD/VDDU/VDDS). Datasheet notes VCCIN must be tied to 5 V.

Clock straps GPIO3/GPIO7 left at their internal pull-up default (11) = 30 MHz crystal, so no strap resistors. GPIO6 default = I2C mode, no SPI ROM.

SATA P1–P3 (3.3 V) are not connected on purpose: P3 is PWDIS on SATA 3.3 drives and a 3.3 V supply there keeps them from spinning up.

Bay activity LED assumed on GPIO0 (bridge firmware default): confirm on the bench; GPIO1 is the alternative.

brain-drain

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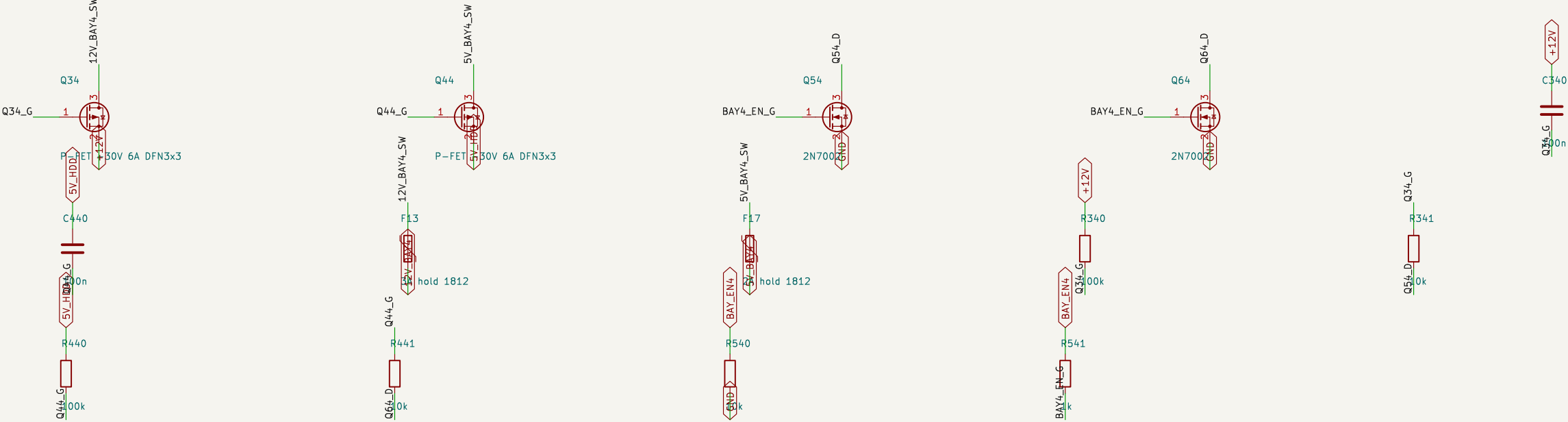
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Rev: v0-generated

KiCad E.D.A. 9.0.8

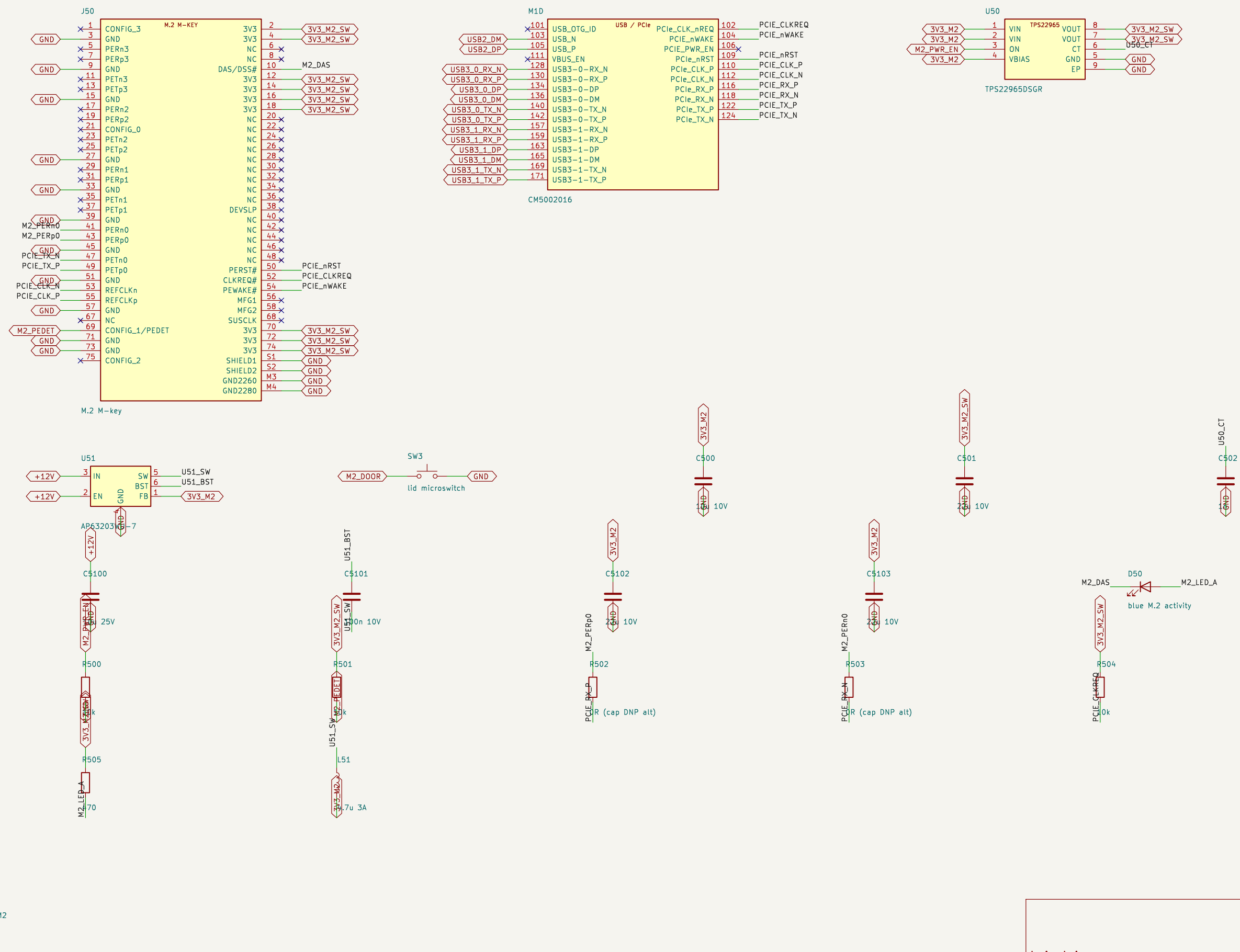
bay-switch-4: Bay 4: switched 12 V and 5 V with soft start and PTC fuses



P-FET high-side switch; 100k gate pull-up holds it off; the 2N7002 pulls the gate down through 10k with 100n gate-source, ~1 ms ramp so the drive's bulk caps do not trip the fuse. BAY_EN pull-down keeps the bay off through bay

brain-drain		
Sheet: /bay-switch-4/ File: bay-switch-4.kicad_sch		
Title: brain-drain - bay-switch-4		
Size: A3	Date:	Rev: v0-generated
KiCad E.D.A. 9.0.8		Id: 14/15

m2-nvme: Bay 5: M.2 M-key on PCIe Gen3 x1, 3V3_M2 buck + load switch, door switch



Polarity: CM5 TX_P -> PETp0 (49), TX_N -> PETn0 (47), PERp0 (43) -> RX_P, PERn0 (41) -> RX_N. The CM5IO reference wires both pairs inverted (TX_P to 47, RX_P to 41), which PCIe link training tolerates; either works.

Only PCIe lane 0 is wired (x1). PET = host transmit (CM5 TX, AC caps on the module). PER = host receive: per the M.2 spec the SSD carries its own TX caps, so R502/R503 are 0R; the same 0402 pads take 100n if a module without caps turns up.

PEDET (pin 69) is pulled up and read by GPIO19: an M.2 SATA module grounds it, and the software refuses the bay instead of trying PCIe.

Door switch SW3 to GPIO4 (software pull-up): closed = low = door shut. Software powers the slot (M2_PWR_EN, GPIO27 → U50) and rescans PCIe.